Invariant PDE

Example: $\Delta f = \lambda f$.

Study what it means that this is "invariant" under coordinate transformations. Certainly it doesn't mean it's "unchanged".

(Note: the connection to coordinate transformations is special to 2nd order Casimir operators.)

Pose $\Delta f = \lambda f$ as an "inactive" **DSolve** problem:

In[2798]:=
```
ΔfCartesian =
 Inactive[DSolve][Laplacian[f[x, y], {x, y}] == λ f[x, y], f[x, y], {x, y}]
```

Out[2798]=
$$\text{DSolve}\left[f^{(0,2)}(x, y) + f^{(2,0)}(x, y) = \lambda\, f(x, y),\, f(x, y),\, \{x, y\}\right]$$

(There is another form of **DSolve** where instead of "**f[x,y]**" you write "**f**", i.e. just the function, not with arguments. This one is a little easier to read, the other one is a little easier to use.)

Mathematica can find one particular solution:

In[2799]:=
```
fExpl = f[x, y] /. Activate[%] // First // Quiet
```

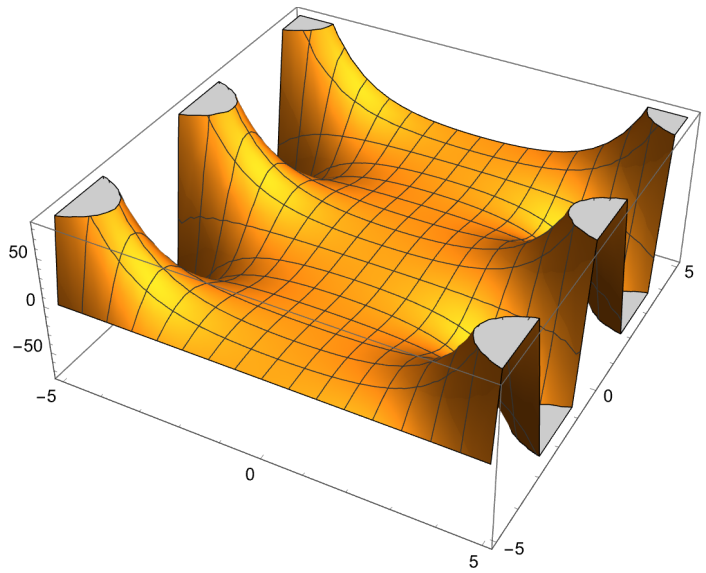
Out[2799]=
$$e^{\sqrt{c_5}\,(-x)}\left(c_1\, e^{2\,\sqrt{c_5}\,x} + c_2\right)\left(c_4 \cos\left(y\,\sqrt{-\lambda + c_5}\right) + c_3 \sin\left(y\,\sqrt{-\lambda + c_5}\right)\right)$$

In[2800]:=
```
% /. {C[_] → 1, λ → -1} // FullSimplify
```

Out[2800]=
$$2\cosh(x)\left(\sin\left(\sqrt{2}\,y\right) + \cos\left(\sqrt{2}\,y\right)\right)$$

In[2801]:=
```
Plot3D[%, {x, -5, 5}, {y, -5, 5}]
```

Out[2801]=



Check that it's indeed a solution with eigenvalue λ:

In[2802]:=
```
Laplacian[fExpl, {x, y}]
% / fExpl  // Simplify
```

Out[2802]=
$$c_5\, e^{\sqrt{c_5}\,(-x)}\left(c_1\, e^{2\,\sqrt{c_5}\,x}+c_2\right)\left(c_4\cos\left(y\,\sqrt{-\lambda+c_5}\right)+c_3\sin\left(y\,\sqrt{-\lambda+c_5}\right)\right)+$$
$$e^{\sqrt{c_5}\,(-x)}\left(c_1\, e^{2\,\sqrt{c_5}\,x}+c_2\right)\left(-c_4\,(-\lambda+c_5)\cos\left(y\,\sqrt{-\lambda+c_5}\right)-c_3\,(-\lambda+c_5)\sin\left(y\,\sqrt{-\lambda+c_5}\right)\right)$$

Out[2803]=
$$\lambda$$

OK. In polar coordinates, the PDE $\Delta f = \lambda f$ reads:

In[2804]:=
```
ΔfPolar =
 DSolveChangeVariables[ΔfCartesian, f, {r, θ}, "Cartesian" → "Polar"] // Simplify
```

Out[2804]=
$$\text{DSolve}\left[\frac{f^{(0,2)}(r,\theta)}{r^2}+\frac{f^{(1,0)}(r,\theta)}{r}+f^{(2,0)}(r,\theta)=\lambda\, f(r,\theta),\, f(r,\theta),\, \{r,\theta\}\right]$$

Here Mathematica used this internal data,
that you know:

In[2805]:=
```
CoordinateTransformData["Polar" → "Cartesian", "Mapping", {r, θ}]
```

Out[2805]=
$$\{r\cos(\theta),\, r\sin(\theta)\}$$

so we can just plug in the change of variables:

In[2806]:=
```
fExplPolar = fExpl /. {x → r Cos[θ], y → r Sin[θ]}
```

Out[2806]=
$$e^{\sqrt{c_5}\,(-r)\cos(\theta)}\left(c_1\, e^{2\,\sqrt{c_5}\,r\cos(\theta)}+c_2\right)\left(c_3\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_4\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)\right)$$

This is the same as what we would get from the built-in version:

In[2807]:=
```
TransformedField["Cartesian" → "Polar", fExpl, {x, y} → {r, θ}]
```

Out[2807]=
$$e^{\sqrt{c_5}\,(-r)\cos(\theta)}\left(c_1\, e^{2\,\sqrt{c_5}\,r\cos(\theta)}+c_2\right)\left(c_3\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_4\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)\right)$$

Now apply the Laplacian Δ to f in polar coordinates:

In[2808]:=
```
Laplacian[fExplPolar , {r, θ}, "Polar"]  // Expand
% / fExplPolar // Simplify
```

Out[2808]=
$$c_1\, c_4\,\lambda\cos^2(\theta)\, e^{\sqrt{c_5}\,r\cos(\theta)}\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_2\, c_4\,\lambda\cos^2(\theta)\, e^{\sqrt{c_5}\,(-r)\cos(\theta)}\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+$$
$$c_1\, c_3\,\lambda\cos^2(\theta)\, e^{\sqrt{c_5}\,r\cos(\theta)}\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_2\, c_3\,\lambda\cos^2(\theta)\, e^{\sqrt{c_5}\,(-r)\cos(\theta)}\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+$$
$$c_1\, c_4\,\lambda\sin^2(\theta)\, e^{\sqrt{c_5}\,r\cos(\theta)}\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_2\, c_4\,\lambda\sin^2(\theta)\, e^{\sqrt{c_5}\,(-r)\cos(\theta)}\cos\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+$$
$$c_1\, c_3\,\lambda\sin^2(\theta)\, e^{\sqrt{c_5}\,r\cos(\theta)}\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)+c_2\, c_3\,\lambda\sin^2(\theta)\, e^{\sqrt{c_5}\,(-r)\cos(\theta)}\sin\left(r\,\sqrt{-\lambda+c_5}\,\sin(\theta)\right)$$

Out[2809]=
$$\lambda$$

OK. It's an eigenfunction, with the same eigenvalue.
(One can just do `Simplify` right away. I did it to show that it takes a calculation.)

[I will add some material here about periodic domains, like a torus, where the basic example is the SL(2) Eisenstein series. If you are interested, see my introductory notes https://tp.hotell.kau.se/marcus/eisenstein (8 pages)]